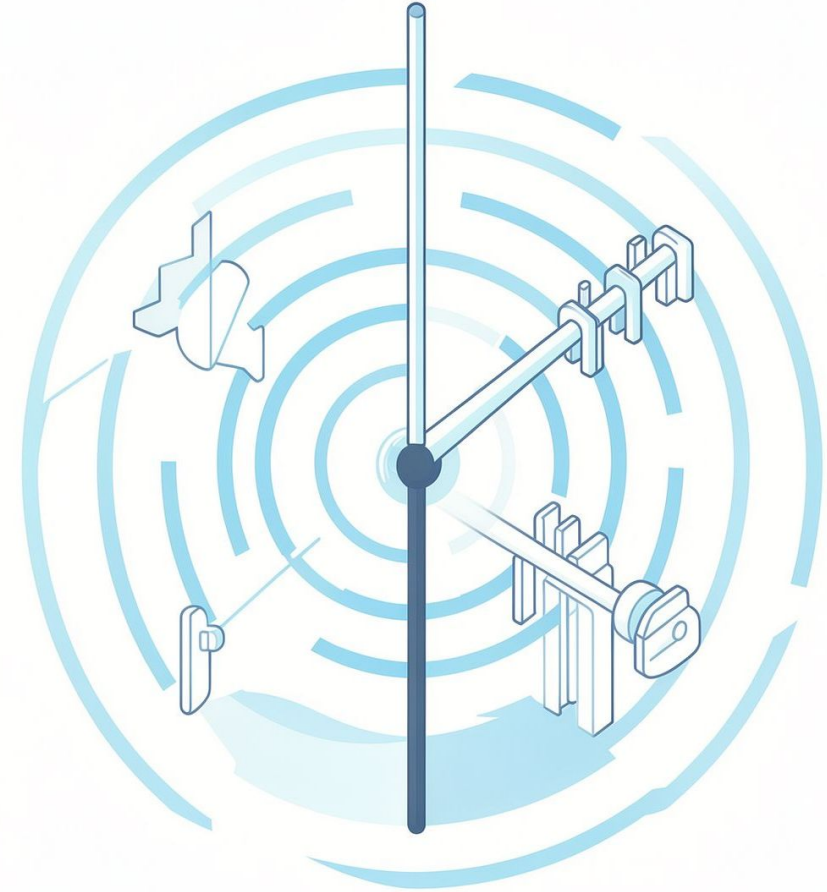


RADAR SIGNAL PROCESSING

Micro-Doppler Simulation of a Maneuvering Quadcopter

Digital Array Radar Signal Processing

1

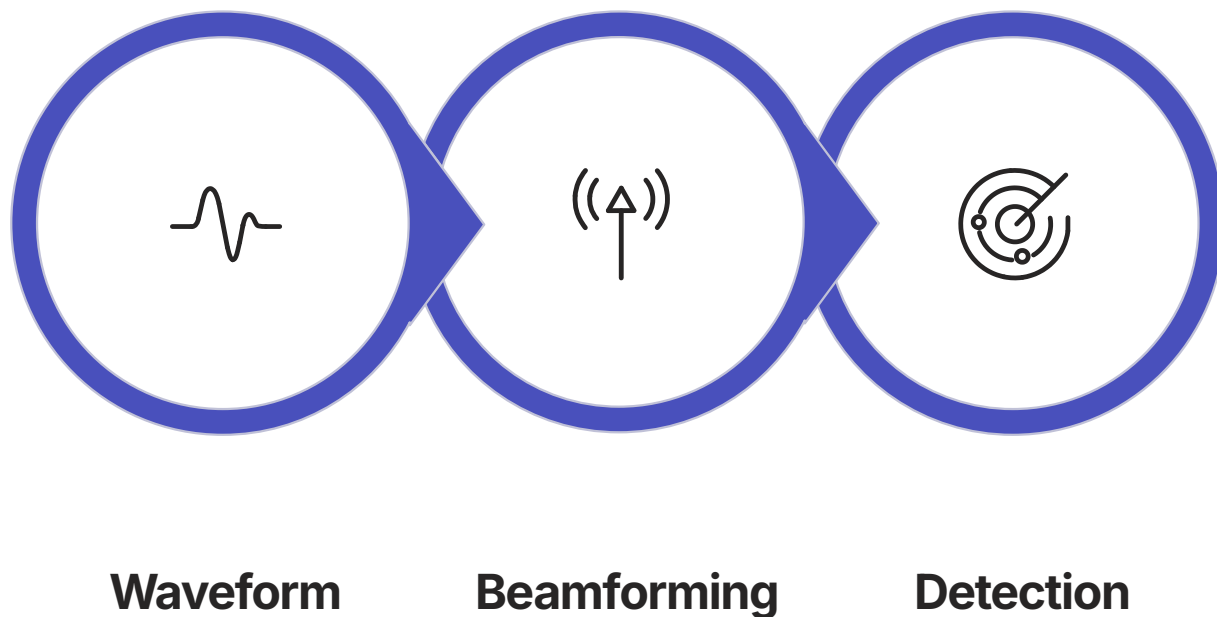


Why Radar-Based UAV Detection?



- Quadcopter generate unique micro-Doppler signatures from rotating propellers
- Distinguishable from birds, vehicles, and other airborne objects
- 5 GHz digital array radar with 4×4 antenna configuration
- Simulates maneuvering drone through three waypoints (A → B → C)

Our approach processes the signal in three distinct stages:



This sequential processing ensures robust and accurate target identification.

System Configuration

Carrier Frequency:

5 GHz

Sampling Frequency:

10 MHz

Pulse Repetition Frequency (PRF):

20 kHz

Waveform:

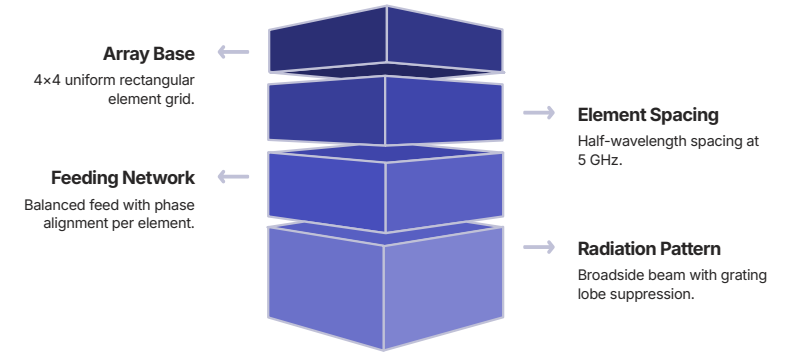
Linear Frequency Modulated (LFM) chirp

Array Configuration:

4×4 Uniform Rectangular Array (URA)

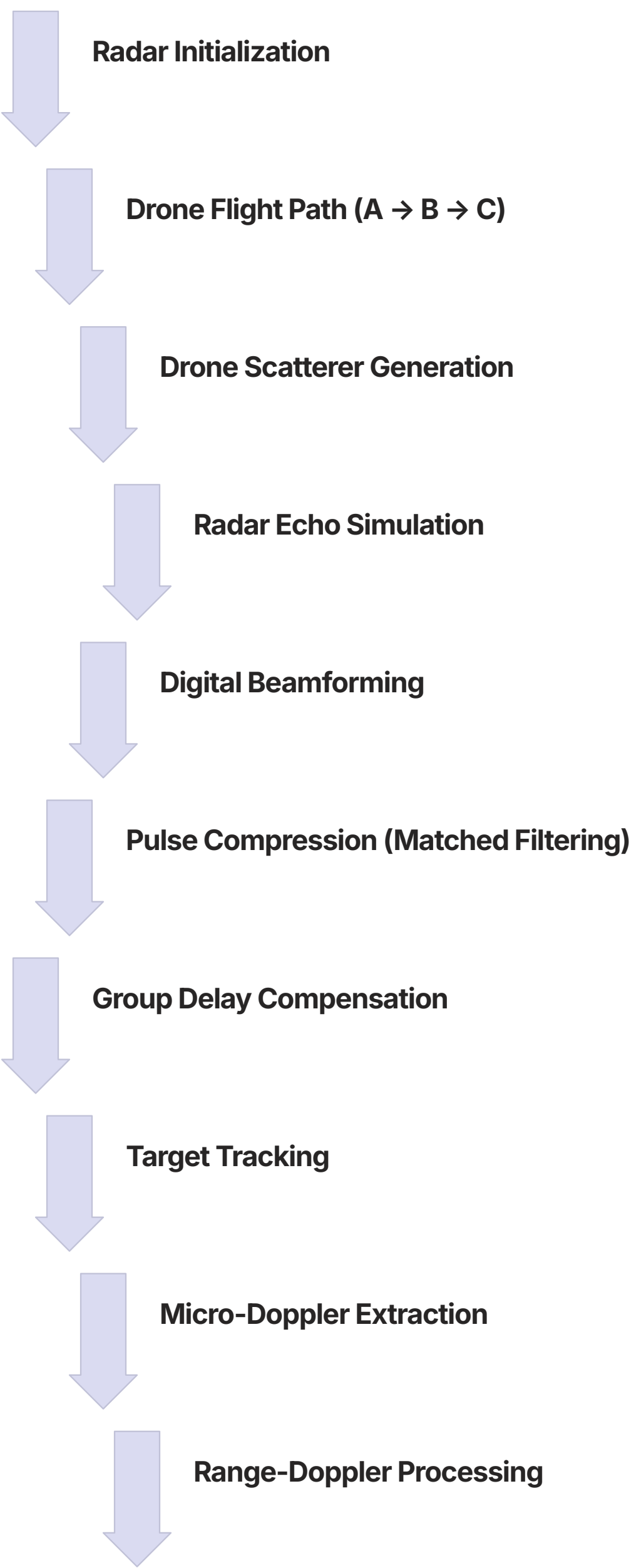
Operating Mode:

Stationary, non-scanning, free-space



The 4x4 Uniform Rectangular Array (URA) provides precise signal reception and transmission capabilities.

10-Step Processing Pipeline



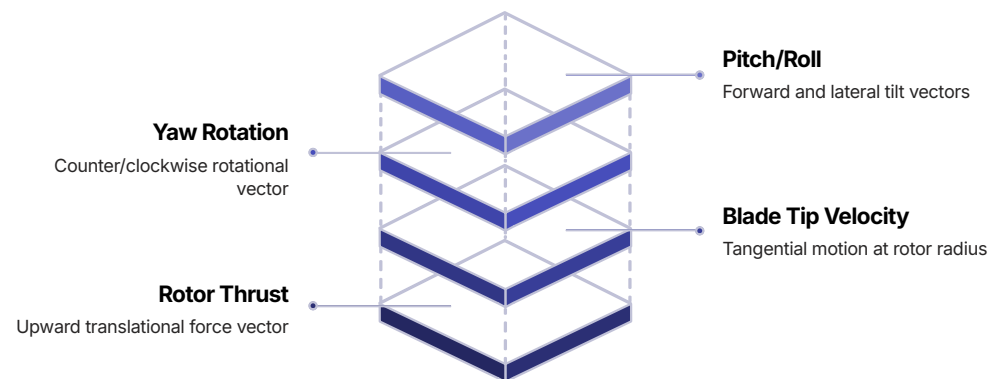
Quadcopter Representation & Motion

Drone Scatterer Model:

- 1 body scatterer
- 8 blade-tip scatterers (2 blades $\times$ 4 rotors)
- Total: 9 radar scatterers

Rotor Geometry:

- Rotor radius: 0.12 m
- 2 blades per rotor
- 4 rotor hubs positioned around body
- Each blade experiences translational + rotational motion



[MATLAB Graph: Drone Trajectory A→B→C]

Core Signal Processing Techniques

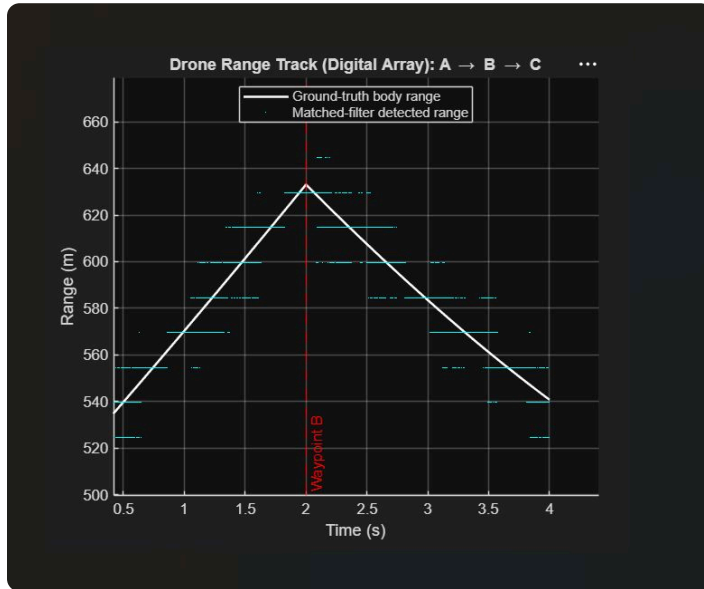
Processing Stage	Algorithm	Purpose
Waveform Generation	LFM Chirp	High range resolution
Pulse Compression	Matched Filtering	Narrow pulse, improved range
Group Delay Correction	Manual shift	Align detected range with true range
Range Processing	FFT-based Range-Doppler	Range profiles & maps
Doppler Processing	128-point FFT	Target radial velocity
Micro-Doppler Analysis	STFT (pspectrum)	Blade-induced signatures
Target Detection	Argmax range-bin	Track strongest return
Digital Beamforming	Delay-and-sum	Coherent array gain

[MATLAB Graph: Algorithm Performance Metrics]

Key Findings

Drone Range Track

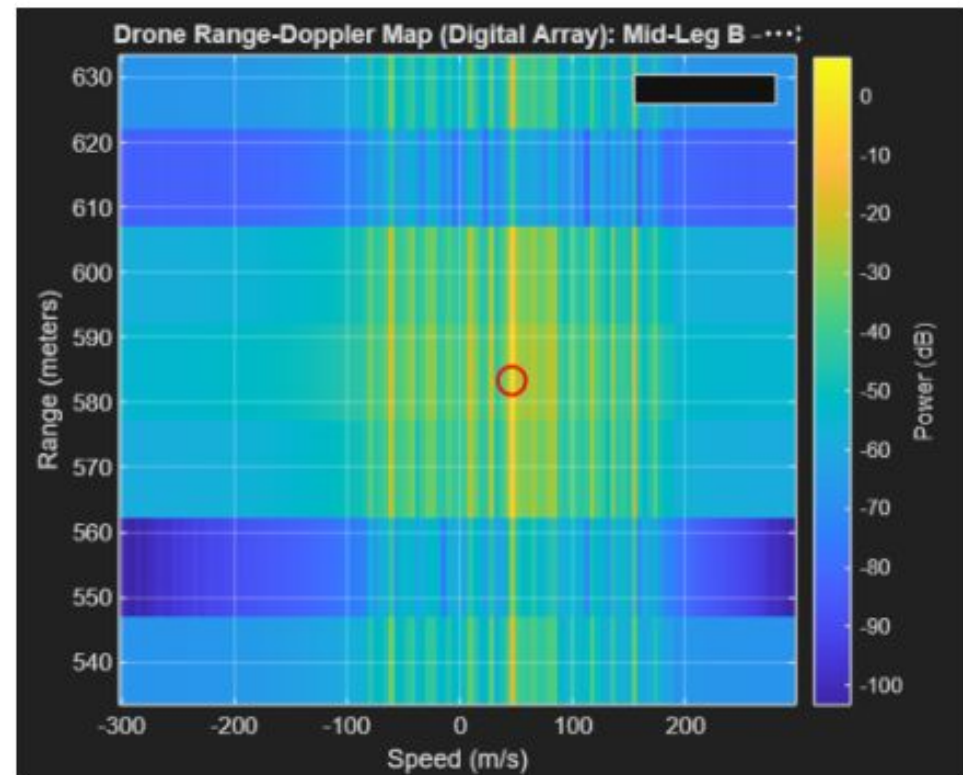
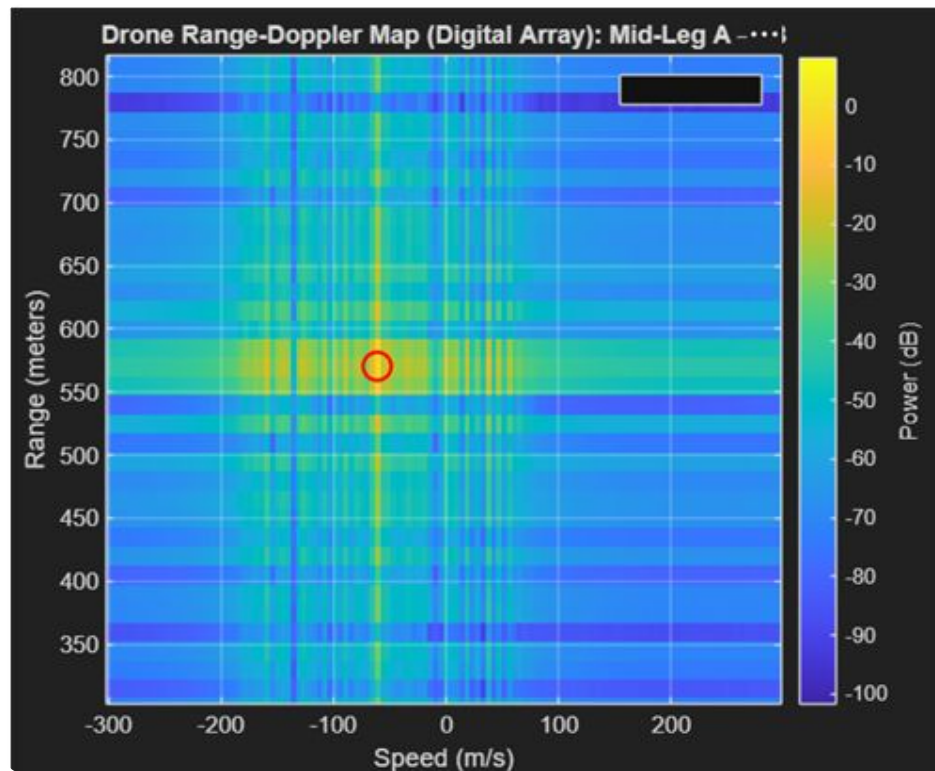
- Detected range closely overlaps ground truth
- Validates accurate range estimation
- Confirms successful target tracking



[MATLAB Graph: Range Track Plot]

Range-Doppler Maps

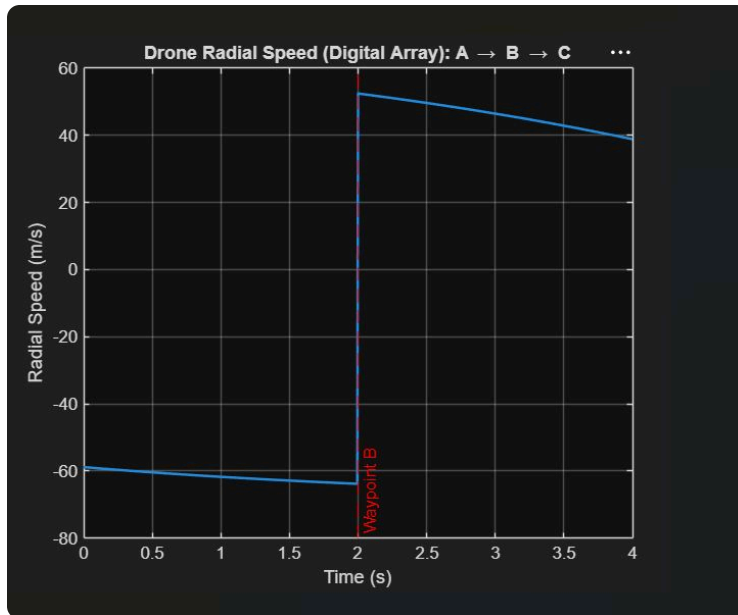
- Mid-leg A→B: Range $\approx 570\text{m}$, Velocity $\approx -60\text{ m/s}$
- Mid-leg B→C: Range $\approx 583\text{m}$, Velocity $\approx +50\text{ m/s}$



[MATLAB Graph: Range-Doppler Maps]

Radial Speed Profile

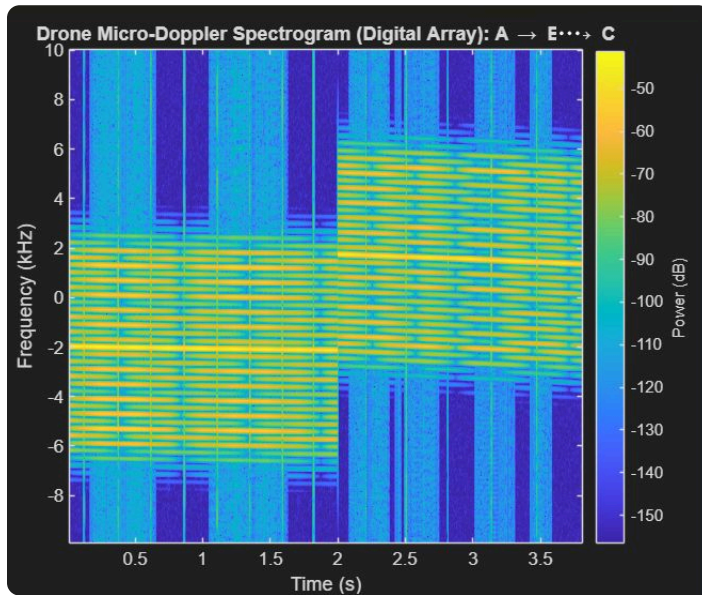
- A→B: Radial velocity ≈ -60 m/s
- B→C: Radial velocity $\approx +50$ m/s
- Confirms waypoint maneuver implementation



[MATLAB Graph: Radial Speed Plot]

Micro-Doppler Spectrogram

- Central band: Drone body Doppler
- Ladder structures: Blade-induced sidebands
- Captures translational + rotational motion



[MATLAB Graph: Micro-Doppler Spectrogram]

Conclusions & Applications

Key Achievements:

- ¹Complete simulation framework validated
- ²Accurate range and velocity estimation
- ³Successful micro-Doppler signature extraction
- ⁴Robust tracking through maneuvering flight
- ⁵Digital beamforming improves SNR and detection

Future Applications:

- ⁶Machine learning-based drone classification
- ⁷Real-time embedded radar systems
- ⁸Multi-target tracking scenarios
- ⁹Atmospheric effects modeling
- ¹⁰Autonomous defense systems

[MATLAB Graph: Classification Performance]

Our extensible platform provides a robust foundation for advanced UAV radar research.